

Appendix D

A Quick Look at Mortgages

This is a very brief introduction to mortgages and mortgage-backed securities. It is intended only to provide a basis for understanding the market dynamics prevalent in the summer of 2003 in which mortgage investors were forced to pay fixed in swaps in order to manage the duration of their portfolios during the massive sell-off in the bond market. For more information on mortgage securities, see Sundaresan (2009) or Fabozzi (2005).

A *mortgage* is a collateralized loan in which the borrower, a homeowner, obtains cash from a lender (typically a bank or mortgage originator) in order to help finance the purchase of the home. The home itself acts as collateral on the loan: in the event of default, the lender takes possession of the house and property.¹

The standard mortgage in the United States is a fixed rate, 30-year loan in which the borrower pays equal payments on a monthly basis for a term of 30 years. It is important to note that a mortgage is a fully self-amortizing loan: the magnitude of each monthly payment is enough to pay off interest due on the loan as well as some principal. Moreover, the monthly payments are “level,” meaning that the payments are the same each month for the 30-year life of the loan. As the life of the mortgage goes by, the outstanding balance on the mortgage decreases at an increasing rate, as a declining proportion

¹One assessment of the quality of a mortgage when regarded as an asset of the lender is known as the *loan to value*, or LTV, which is the ratio of the loan size to the value of the property. The smaller the LTV, the higher the credit quality of the mortgage.

Month	Beginning Balance	Interest on Outstanding Balance	Principal Paid Down	Ending Balance
1	\$100,000.00	\$500.00	\$99.55	\$99,900.45
2	99,900.45	499.50	100.05	99,800.40
3	99,800.40	499.00	100.55	99,699.85
4	99,699.85	498.50	101.05	99,598.80
5	99,598.80	497.99	101.56	99,497.24
⋮	⋮	⋮	⋮	⋮
358	1,780.81	8.90	590.65	1,190.17
359	1,190.17	5.95	593.60	596.57
360	596.57	2.98	596.57	0.00

Table D.1: Principal and Interest Payments on a \$100,000 30-Year 6% Mortgage

of the fixed monthly payment needs to be applied to paying off interest and is instead applied to paying off principal.

Consider, for example, a 30-year loan of \$100,000 on which the annual mortgage rate is 6%. One can easily calculate on a financial calculator that the monthly mortgage payment is \$599.55. Alternatively, the monthly mortgage payment may be calculated using the formula

$$P \times \frac{y/m}{1 - (1 + y/m)^{-mn}},$$

where P is the face of the loan, y is the mortgage rate, m is the number of payments per year, and n is the number of years of the loan.²

In the first month, the interest that needs to be paid on the loan is $\$100,000 \times 6\%/12 = \500 , and the remaining \$99.55 portion of the monthly payment reduces the outstanding principal balance accordingly. By the last month of the mortgage, the outstanding principal balance is only \$596.57. The final payment pays off interest due of \$2.98, and the remainder of the monthly payment is enough to pay off the balance. Table D.1 demonstrates what happens to the principal balance and interest paid over the life of the loan.

²For our purposes here, we will sometimes round calculations of the monthly mortgage payment to the nearest cent. This will introduce a slight rounding error in the principal balance that can safely be ignored.

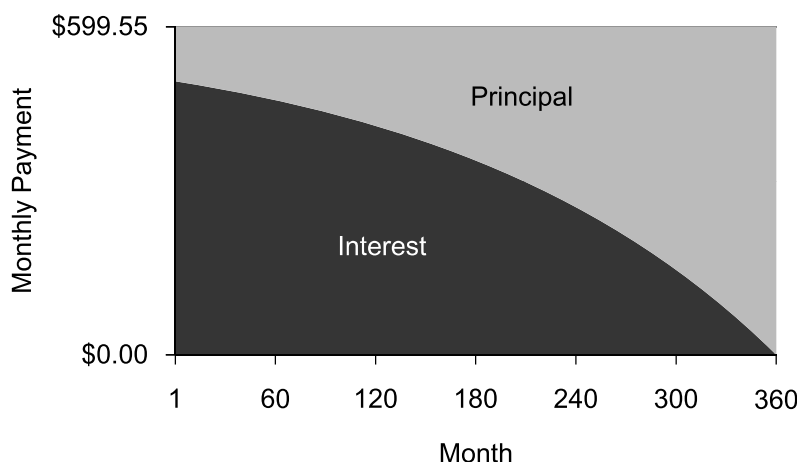


Figure D.1: Principal and Interest Payments Over the Life of a Mortgage

Over the life of the loan, the interest component of the monthly payment steadily declines as more and more principal is paid off with each passing month. Notice that the sum of the interest paid on the outstanding balance and the principal paid down (the sum of the third and fourth columns, respectively, in Table D.1) is always a constant \$599.55. Figure D.1 shows graphically how the division of principal and interest varies over the life of the loan.

One of the most salient features of most mortgages is that the borrower has the right to prepay the mortgage at any time over the life of the loan.³ That is, the borrower may at any time pay off the outstanding principal balance. The borrower may choose to do so for any of a number of reasons: his job may necessitate that he move to another city and sell his house, he may decide to sell his current home to buy a bigger one or a smaller one and so on. But perhaps the most interesting reason that a homeowner may prepay his existing mortgage is due to the possible economic incentive to *refinance* his loan, which will arise if mortgage rates go lower.

³Mortgages are typically *nontransferable*, meaning that if a homeowner moves from his house, he cannot apply his existing mortgage to his new home, nor can he in any way assign it to the new owner of his old house. An exception is Federal Housing Administration (FHA) mortgages, which may, under certain conditions, be assumed by another qualified borrower upon sale of the home.

For example, suppose that after the homeowner takes out the \$100,000 30-year mortgage at 6%, mortgage rates drop to 5.75%. Instead of making payments of \$599.55 as per the loan at 6%, the homeowner could immediately prepay the loan and instead take out a new 30-year mortgage at a rate of 5.75%, which works out to be a monthly payment of \$583.57. That means that over the life of the loan the borrower will save $(\$599.55 - \$583.57) \times 360 = \$5,752.80$ in nominal interest expense if he were to immediately refinance.⁴ The borrower may or may not choose to refinance: by refinancing, he will incur transaction costs (legal and other closing costs) that will diminish his economic incentive to refinance. Or perhaps the borrower is not that savvy and does not think to refinance in a lower rate environment. Other times a borrower who met the underwriting standards necessary to qualify for his original loan may no longer qualify for a new loan by the time rates fall (if, e.g., the value of the home, which serves as collateral for the loan, has fallen too significantly relative to the size of the loan). But one thing is true: the lower rates go after he takes out his initial loan, the greater his incentive to prepay the mortgage and refinance at a lower rate.

Let's take a look at this mortgage now from the perspective of the lending institution. The lending institution owns a (risky) asset that pays fixed coupons each month, and has the possibility of being prepaid as rates fall.⁵ In some sense, then, this asset resembles a callable bond: as rates fall, the asset may be "called away" by the "issuer," who in this case is of course the homeowner. Let's assume that as rates do fall, the probability that the loan is prepaid increases.⁶ Then as rates fall, the probability that the lender receives his principal back early increases. Conversely, as rates rise, the probability

⁴Note that in a 30-year 6% loan the borrower makes total payments of \$215,838.19, of which \$100,000 goes to paying back principal and the remaining \$115,838.19 goes to paying off interest expense. In the 30-year 5.75% loan, the borrower makes total payments of \$210,086.23, of which \$100,000 goes to paying back principal and the remaining \$110,086.23 goes to paying off interest expense. This simplistic example ignores considerations such as the timing of cash flows, transaction costs to refinance, and so on, but nonetheless demonstrates the point that as mortgage rates go lower, the borrower will have an incentive to refinance.

⁵The asset is risky not only in terms of the uncertainty of the cash flows that the loan may generate but also in terms of the possibility of the borrower defaulting.

⁶This may not necessarily be the case. The analysis of the possible prepayment of mortgages is a very complicated area in finance that takes into consideration many variables, including the credit (FICO) score of a borrower, the loan size, current LTV, and geography. Another variable that is often considered is *burnout*, which refers to the phenomenon in which borrowers who have previously not taken advantage of a lower rate environment to refinance are not likely to do so in a subsequent lower rate environment. If this is the case, then a low rate environment may not necessarily increase the probability of prepayment.

that the lender's asset will remain outstanding until the final maturity of the loan increases.

Oftentimes, lenders such as banks and mortgage originators do not wish to hold the mortgages they originate on their books. These institutions make money through the *origination* of loans, the process by which they lend money to new borrowers. If they sell their mortgages after they are originated, they will have more balance sheet available to make additional loans.⁷ This works out well because there are investors in the market who are interested in buying their mortgages, usually on a securitized basis. Buying mortgages loan by loan on an unsecuritized basis would prove difficult for most potential investors, as investors generally don't have the resources to dedicate to analyzing individual loans, which contain credit risk due to the nature of the individual borrowers. Some of the GSEs—namely, Fannie Mae and Freddie Mac, as well as Ginnie Mae (as a direct arm of the government)—help to facilitate the process of making mortgages available to investors by

- Buying individual mortgages from lending institutions;
- Bundling individual mortgages with similar characteristics into a larger grouping known as a *pool*;⁸
- Providing a guarantee on the mortgages in the pool, thereby protecting investors who buy the mortgages from the possibility of default of individual loans in the pool;
- Securitizing the pool and making it available for sale to investors on a liquid basis.

Investors who buy these securitized mortgage pools, known as *agency mortgage-backed securities*, typically earn a higher yield than they would on comparable duration Treasuries, even though they have the government's imprimatur against default risk. The reason for the higher yield is the prepayment risk that the investor faces: as the market rallies, borrowers in the pool are more likely to prepay their loans, and therefore the weighted average life of the security decreases. Conversely, as the market sells off, borrowers

⁷There are times, however, when these institutions choose to maintain a portion of the loans they originate on their balance sheets. If, for example, a bank believes other asset classes it normally invests in are less attractive than usual and the commercial lending activities of the bank are for some reason curtailed, the bank may find it advantageous to retain a portion of the mortgages it originates in its own portfolio.

⁸The reason that the agencies bundle mortgages with similar characteristics (e.g., coupon, maturity, loan size) is that it makes for a pool that has uniformly similar characteristics and is thus easier for investors to analyze on a consistent basis.

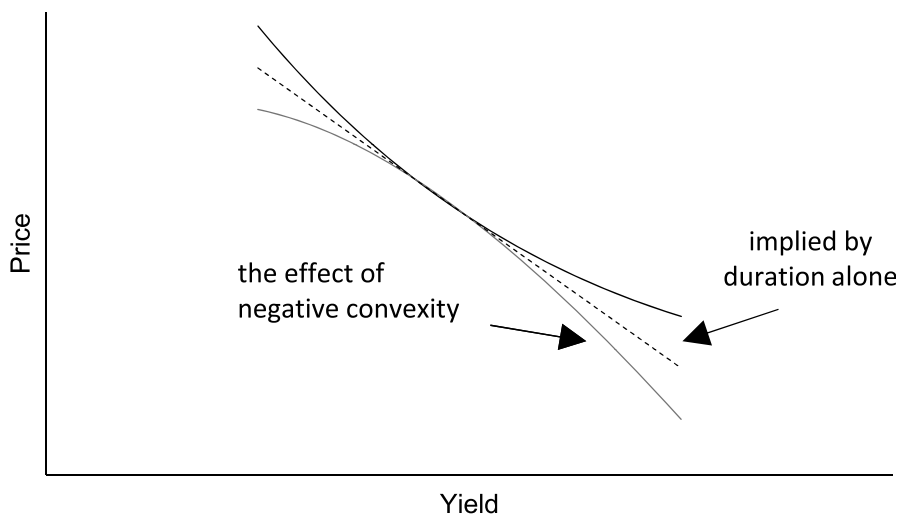


Figure D.2: The Negative Convexity of Mortgages (*lower curve*) Versus a Security with Positive Convexity (*higher curve*)

in the pool are less likely to prepay their loans, and therefore the weighted average life of the security increases. This does imply the following very important characteristic of MBSs: *as the market rallies, the duration of MBSs decreases, and as the market sells off, the duration of MBSs increases.*

Now let's take a step back for a minute: suppose we are a fixed income investor, and we expect the market to rally. Given our view, we would prefer to own assets that have a long duration rather than a short duration. The reason for this is that longer duration assets, by definition, have greater sensitivity to changes in interest rates. And if rates are going down, we want a lot of price sensitivity to benefit from greater price appreciation of the assets in our portfolio. If we instead own short duration assets, these would have limited price sensitivity, and their value would not increase as dramatically in a rally. On the other hand, if we expect that the market will sell off, we would prefer to hold short duration assets in our portfolio. A sell-off in the market will diminish the value of our fixed rate bonds. Those with a short duration will depreciate in value less than those with longer durations.

Unfortunately, this is anything but the experience we will have if we own MBSs. As the market rallies (and we wished that we owned long duration assets to maximize price appreciation in our fixed income portfolio), borrowers in the pool will begin to prepay their loans as the economic incentive to do so increases. This will shorten the duration of the MBSs we own, with

principal being returned to the bondholder at par instead of allowing us to continue to own a bond that is now paying a premium coupon. Conversely, as the market sells off (and we are hoping to own short duration assets to minimize the price depreciation of our portfolio), the incentive for borrowers to prepay will decrease, and thus the expectation that prepayments will occur will decline. This will cause the duration of MBSs to increase. In sum, this is a bad thing: as the market rallies, and we want our mortgages to stick around, they disappear, and when the market sells off, and we want our mortgages to disappear, they stick around. More formally, this phenomenon is characteristic of the *negative convexity* inherent in MBSs. The effect of negative convexity on the price performance of MBSs is demonstrated in Figure D.2.

This negative convexity is the manifestation of the options that the MBS bondholder is inherently short to the homeowners. Of course, investors who buy MBSs are compensated for enduring this negative convexity: the coupons paid by MBSs issued at par are higher than on comparable duration Treasuries issued at par. The excess yield of MBSs more than compensates for small market moves and undercompensates for larger moves.

Let's now refer back to the market dynamics of the summer of 2003 in which the market sold off dramatically, nearly 150 bps in about six weeks. Investors who held large portfolios of mortgages found that the durations of their portfolios were increasing (or put another way, their mortgage portfolios were *extending*). In order to manage the extension of their mortgage portfolios, such investors found it necessary to enter into trades to offset the increasing duration of their portfolios. One way they could do this was by selling Treasuries, and another way they could do this was by paying fixed in swaps.⁹ As large mortgage investors increased their hedging activity in both the Treasury and especially the swaps market, this contributed to and exacerbated the huge sell-off and spread widening that was discussed in Section 2.2.1.

⁹Note that as the market is selling off (with yields rising and bond prices falling), investors who need to hedge their mortgages are being put into the position of having to sell Treasuries at lower prices or pay a higher fixed rate in swaps than they could have before the sell-off. Either of these alternatives are disadvantageous and are characteristic of having to hedge the negative convexity implicit in mortgage securities.

Appendix E

The Normal Model

Here is a heuristic sketch of the derivation of the value of swaptions under the assumption of a normal distribution, as compared to the standard Black-Scholes assumption of lognormality. We employ risk-neutrality. In particular, let us consider pricing a T into n payer swaption struck at X , under the assumption that the underlying swap rate $R \sim N(F, \sigma_N^2 T)$, where F denotes the n -year swap rate, T years forward. In this formulation, we refer to σ_N as *normal vol*. For ease of notation, define $\tau = \sigma_N \sqrt{T}$.

For simplicity, assume the swap pays annually on the fixed side and consider now valuing the cash flow at time t_i associated with exercise at time T , where $t_i = T + 1, T + 2, \dots, T + n$. The future value of this cash flow at time t_i is equal to

$$\begin{aligned} & \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tau}} \exp \left[-\frac{(R-F)^2}{2\tau^2} \right] \max(0, R-X) dR \\ &= \int_{-\infty}^X \frac{1}{\sqrt{2\pi\tau}} \exp \left[-\frac{(R-F)^2}{2\tau^2} \right] \max(0, R-X) dR \\ & \quad + \int_X^{\infty} \frac{1}{\sqrt{2\pi\tau}} \exp \left[-\frac{(R-F)^2}{2\tau^2} \right] \max(0, R-X) dR \\ &= \int_X^{\infty} \frac{1}{\sqrt{2\pi\tau}} \exp \left[-\frac{(R-F)^2}{2\tau^2} \right] (R-X) dR. \end{aligned} \tag{E.1}$$

Let us now make a change of variables and let $x = (R-F)/\tau$. Notice that with this substitution we have $dR = \tau dx$. Substituting into Equation E.1 gives